

5.3.14 I/O port characteristics

General input/output characteristics

Unless otherwise specified, the parameters given in Table 46 are derived from tests performed under the conditions summarized in Table 16. All I/Os are designed as CMOS and TTL-compliant.

Note: For information on GPIO configuration, refer to the application note *STM32 GPIO configuration for hardware settings and low-power consumption (AN4899)*.

Table 46. I/O static characteristics

The I/O structure options listed in this table can be a concatenation of options including the option explicitly listed. For instance, TT_a refers to any TT I/O with _a option. TT_xx refers to any TT I/O and FT_xx refers to any FT I/O. All I/Os are CMOS- and TTL-compliant (no software configuration required). Their characteristics cover more than the strict CMOS-technology or TTL parameters. The coverage of these requirements is shown in the figure below. The minimum and maximum values are specified for a junction temperature (T_J) of 125°C.

Symbol	Parameter	Condition	Min	Typ	Max	Unit
V_{IL}	I/O input low level voltage	$2.7\text{ V} < V_{DDIOx} < 3.6\text{ V}$	-	-	$0.3V_{DD}^{(1)}$	V
	I/O input low level voltage		-	-	$0.4 V_{DD} - 0.1^{(2)}$	
V_{IH}	I/O input high level voltage	$2.7\text{ V} < V_{DDIOx} < 3.6\text{ V}$	$0.7V_{DD}^{(1)}$	-	-	V
	I/O input high level voltage		$0.52V_{DD} + 0.18^{(2)}$	-	-	
$V_{HYS}^{(2)}$	TT_xx, FT_xxx and NRST I/O input hysteresis	$2.7\text{ V} < V_{DDIOx} < 3.6\text{ V}$	-	300	-	mV
$I_{leak}^{(3)}$	FT_xx Input leakage current ⁽²⁾	$0 < V_{IN} \leq \text{Max}(V_{DDXXX})^{(4)}$	-	-	± 200	nA
		$\text{Max}(V_{DDXXX}) < V_{IN} \leq \text{Max}(V_{DDXXX} + 1\text{ V})^{(5)(6)(4)}$	-	-	± 2500	
		$\text{Max}(V_{DDXXX} + 1) < V_{IN} \leq 5.5\text{ V}^{(5)(6)(4)}$	-	-	750	
	TT_xx Input leakage current	$0 < V_{IN} \leq \text{Max}(V_{DDXXX} + 1\text{ V})^{(4)}$	-	-	± 200	
R_{PU}	Weak pull-up equivalent resistor ⁽⁷⁾	$V_{IN} = V_{SS}$	30	40	50	k Ω
R_{PD}	Weak pull-down equivalent resistor ⁽⁷⁾	$V_{IN} = V_{DD}$	30	40	50	
C_{IO}	I/O pin capacitance	-	-	5	-	pF

- Compliant with CMOS requirements.
- Specified by design - Not tested in production.
- This parameter represents the pad leakage of the I/O itself. The total product pad leakage is provided by the following formula:
 $I_{Total_leak_max} = 10\text{ }\mu\text{A} + [\text{number of I/Os where } V_{IN} \text{ is applied on the pad}] \times I_{lkg\text{ max}}$
- $\text{Max}(V_{DDXXX})$ is the maximum value of all the I/O supplies.
- To sustain a voltage higher than the minimum of V_{DD} or V_{DDA} plus 0.3 V, disable the internal pull-up and pull-down resistors.
- V_{IN} must be less than $\text{Max}(V_{DDXXX}) + 3.6\text{ V}$.
- The pull-up and pull-down resistors are designed with a true resistance in series with a switchable PMOS/NMOS. This PMOS/NMOS contribution to the series resistance is minimal (~10% order).

All I/Os are CMOS- and TTL-compliant (no software configuration required). Their characteristics cover more than the strict CMOS-technology or TTL parameters. The coverage of these requirements is shown in the following figure.